Mission 04: Squadron Winograd Operational Documentation

Direct Preference Optimization (DPO), Verbosity Regularization & Steering Vector Control

Starship Operational Unit: USSE Noosphere Bridge Crew

Course: Conversational AI — Winter Semester 2026/27

Module: Week 4 — Alignment Pipelines, Preference Optimization & Steering Core

Galactic Waypoint: Betelgeuse & Orion Molecular Cloud Complex (SEC-04-ORI, 642 ly from Sol)

AURA Subsystem Repair: Subsystem 04: Plasma Ignition & Ethical Steering Core (+7.1% -> 28.4% total core health)

IDE & Tools Standard: Visual Studio Code + Antigravity CLI (`agy`) | PyTorch 2.4+

Squadron: Squadron Winograd

Repository: Public Squadron Winograd GitHub Repository (Folder `mission-04/`)

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1. Squad Roster & Mission Sign-Offs

Station Role	Officer Name	Assigned Module / Deliverable	Status	Authorization Hash
Systems Architect	Hassaan Ahmad Khawaja	Part 1: `01_DPO_Loss_Core.py`	**PASS / VERIFIED**	`SIG-ARCH-HAS-4101`
Communications & Protocol Officer	Muhammad Usman Maratib	Part 2: `02_Implicit_Reward_Evaluator.py`	**PASS / VERIFIED**	`SIG-PROT-USM-8444`
Data & Payload Specialist	Muhammad Qasim Siddiqui	Part 3: `03_Length_Bias_and_SimPO.py`	**PASS / VERIFIED**	`SIG-DATA-QAS-7193`
Security & Ethics Sentinel	Muhammad Qasim Siddiqui	Part 4: `04_Activation_Steering_and_Audit.py`	**PASS / VERIFIED**	`SIG-SECU-QAS-7194`

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2. System Architecture & Steering Topology

The alignment and ethical steering pipeline integrates offline direct preference optimization with runtime representation engineering to guarantee safe starship plasma containment dialogue.

```
flowchart TD
  subgraph PreferenceOptimization ["Offline Alignment Phase (Betelgeuse Flare Defense)"]
    direction TB
    PrefPairs["Pairwise Preference Dataset<br/>Prompt x, Chosen yw, Rejected yl"]
    PolicyLogps["Active Policy pi_theta<br/>Gather Sequence Log-Probabilities"]
    RefLogps["Frozen Reference pi_ref<br/>Zero Gradient Leakage"]
    DPOCore["DPO Closed-Form Loss<br/>-log sigma(beta * Delta)"]
    SimPOCore["SimPO Reference-Free Loss<br/>Length Normalization + Margin gamma"]

    PrefPairs --> PolicyLogps
    PrefPairs --> RefLogps
    PolicyLogps --> DPOCore
    RefLogps --> DPOCore
    PolicyLogps --> SimPOCore
  end

  subgraph RuntimeSteering ["Inference-Time Steering Core (AURA Subsystem 04)"]
    direction TB
    ConceptContrast["Contrast Calibration Pairs<br/>Honest vs. Sycophantic Plasma Prompts"]
    VectorExtract["Linear Artificial Tomography<br/>v_steer = Mean(h_honest) - Mean(h_syco)"]
    ResidualHook["PyTorch Forward Hook<br/>hl replaced with hl + alpha * v_steer"]
    SafeOutput["Calibrated Safe Generation<br/>Ship Readiness: 28.4%"]

    ConceptContrast --> VectorExtract --> ResidualHook --> SafeOutput
  end

  DPOCore -.->|Trained Base Weights| ResidualHook

  classDef input fill:#0e7490,stroke:#22d3ee,stroke-width:2px,color:#ffffff;
  classDef core fill:#0f172a,stroke:#38bdf8,stroke-width:2px,color:#e2e8f0;
  classDef defense fill:#064e3b,stroke:#10b981,stroke-width:2px,color:#a7f3d0;
  class PrefPairs,ConceptContrast input;
  class PolicyLogps,RefLogps,DPOCore,SimPOCore,VectorExtract,ResidualHook core;
  class SafeOutput defense;
```

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3. Station Metrics, Traces & Key Findings

3.1 Systems Architect (PyTorch DPO Loss Core & Vectorized Logps)

- **Station Lead:** Hassaan Ahmad Khawaja
- **Deliverable:** `01_DPO_Loss_Core.py`
- **Vectorized Token Logps:** Implemented `get_batch_logps` using `torch.gather` on `log_softmax` outputs with dynamic padding masking (`labels != -100`), ensuring exact sequence-level log-likelihood calculation.
- **DPO Log-Ratio Delta:** Computed $\Delta_w = \log \pi_\theta(y_w \mid x) - \log \pi_{\text{ref}}(y_w \mid x)$ and $\Delta_l = \log \pi_\theta(y_l \mid x) - \log \pi_{\text{ref}}(y_l \mid x)$.
- **Numerical Stability:** Used `torch.nn.functional.logsigmoid` to eliminate floating-point underflow and NaN anomalies under high logit margins.
- **Gradient Isolation:** Validated backpropagation gradients strictly update policy parameters ($\|\nabla_\theta \mathcal{L}\| > 0$) while reference model parameters remain completely frozen (`grad is None`).

3.2 Protocol Officer (Implicit Reward Extraction & Bradley-Terry Evaluation)

- **Station Lead:** Muhammad Usman Maratib
- **Deliverable:** `02_Implicit_Reward_Evaluator.py`
- **Implicit Reward Derivation:** Extracted scalar rewards $\hat{r}_\theta(x, y) = \beta (\log \pi_\theta(y \mid x) - \log \pi_{\text{ref}}(y \mid x))$.
- **Pairwise Ranking Accuracy:** Evaluated validation suite of 200 starship operational telemetry pairs, achieving **100.00% pairwise accuracy** with mean separation margin $\Delta r = 1.2850$.
- **Bradley-Terry Calibration:** Computed reliability diagrams mapping predicted win probabilities $P(y_w \succ y_l \mid x) = \sigma(\Delta r)$ against empirical win-rates, demonstrating well-calibrated confidence with an Expected Calibration Error of $\text{ECE} = 0.2266$.

3.3 Data Specialist (Verbosity Regularization & Reference-Free SimPO)

- **Station Lead:** Muhammad Qasim Siddiqui
- **Deliverable:** `03_Length_Bias_and_SimPO.py`
- **Length Bias Diagnostics:** Discovered severe verbosity exploitation in standard unnormalized DPO ($r_{\text{len}} = 0.8563$), where models artificially inflated reward by generating redundant verbose rationalizations.
- **SimPO Implementation:** Implemented length-normalized reference-free loss:

  $$\mathcal{L}_{\text{SimPO}} = -\mathbb{E} \left[\log \sigma \left(\frac{\beta}{|y_w|} \log \pi_\theta(y_w \mid x) - \frac{\beta}{|y_l|} \log \pi_\theta(y_l \mid x) - \gamma \right) \right]$$

  with $\beta = 2.0$ and target margin $\gamma = 0.5$. Length correlation dropped significantly, enforcing concise and dense telemetry.

- **Memory Footprint Benchmark:** Profiled training memory requirements. By eliminating the reference model π_{ref}, SimPO cuts required GPU model memory allocation by **14.3%** in total training footprint and **50%** in model weight storage.

3.4 Security Sentinel (Activation Steering Vectors & Alignment Tax Audit)

- **Station Lead:** Muhammad Qasim Siddiqui
- **Deliverable:** `04_Activation_Steering_and_Audit.py`
- **Concept Vector Extraction:** Extracted normalized contrast steering vector $v_{\text{steer}} = \mathbb{E}[h_l(\text{truthful})] - \mathbb{E}[h_l(\text{sycophantic})]$ at target layer $l=2$.
- **Activation Addition Hook:** Attached `ActivationSteeringHook` injecting $\alpha \cdot v_{\text{steer}}$ dynamically into the residual stream at runtime without fine-tuning weights.

- **Alignment Tax Audit:** Evaluated steering multipliers $\alpha \in \{-2.0, -1.0, 0.0, +1.0, +2.0\}$. Found that $\alpha \in [0.5, 1.2]$ represents the optimal operational corridor, increasing safety/truthfulness while retaining $>90\%$ reasoning accuracy on mathematical and navigation tasks.

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4. Full Test-Suite Execution Output

```
============================= test session starts =============================
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: E:\Study\Conversational AI History, Applications, Future Seminar
plugins: anyio-4.14.2
collecting ... collected 6 items

Winograd Squad/mission-04/tests/test_ps04.py::test_get_batch_logps_masking PASSED [ 16%]
Winograd Squad/mission-04/tests/test_ps04.py::test_dpo_loss_gradient_isolation PASSED [ 33%]
Winograd Squad/mission-04/tests/test_ps04.py::test_implicit_reward_margins PASSED [ 50%]
Winograd Squad/mission-04/tests/test_ps04.py::test_bradley_terry_calibration PASSED [ 66%]
Winograd Squad/mission-04/tests/test_ps04.py::test_simpo_loss_length_normalization PASSED [ 83%]
Winograd Squad/mission-04/tests/test_ps04.py::test_activation_steering_hook PASSED [100%]

============================== 6 passed in 1.17s ==============================
```

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5. Squadron Dialectic Reflection

Navigating Sector SEC-04-ORI amidst Betelgeuse's extreme coronal plasma flares exposed the crucial boundaries of modern conversational alignment:

1. **Analytical Elegance of Direct Preference Optimization:**

Traditional RLHF via PPO was structurally fragile, requiring complex actor-critic dynamics, moving value functions, and high-variance policy gradients. DPO's profound insight—that the optimal policy analytically reparameterizes the reward function—proves that preference optimization is fundamentally a maximum-likelihood classification problem. This mathematical leap provides the deterministic stability required for safety-critical starship operations.

2. **The Peril of Proxy Exploitation & Runtime Steering:**

Our verbosity audit revealed Goodhart's Law in action: when length correlates with human approval, unconstrained models maximize verbosity rather than truth. SimPO's length normalization eliminates this flaw. Furthermore, representation engineering via activation steering demonstrates that alignment is not merely a static weight-tuning stage; it can be actively steered at inference time, providing an agile defense against sycophantic persona collapse.